

Erdos #319 - Q-Ledger Addendum v0.2

Update to the prior progress report after the final 5-6 messages in Doc.pdf

Purpose. This addendum updates the earlier Erdos #319 progress package. It is not a replacement for the full progress report. It records the new status after the final messages in the uploaded transcript, especially the verifiable formulation on fixed Q, arbitrary fixed Q, the sigma_Q joint gate, and the cross-number correction.

Headline. We have not solved Erdos #319. We have, however, made serious fixed-Q progress. The arbitrary fixed-Q problem is now closer because it has been reduced to a precise target-selection / finite zero-sum / cross-number obstruction problem. Growing Q_N and the full problem remain long-term.

1. Delta from the previous report

1. The Phi_q gate is a mostly elementary/checkable analytic inequality.
2. The fixed-Q singleton-gate theorem candidate is still not fully verified, but its proof architecture is clean.
3. The arbitrary fixed-Q problem is now expressed as a target-selection problem over $T_Q = \{R(Q') : \text{empty} \neq Q' \subseteq Q\}$.
4. A dynamic single-prime gate lemma candidate was isolated.
5. A joint-avoidance / sigma_Q gate appears to be a genuinely provable fixed-Q lemma.
6. Prime-only fixed Q is likely close, conditional on the Liu-Sawhney reachability interface.
7. Composite/mixed fixed Q reduces to a finite target-selection obstruction.
8. The obstruction has a natural finite abelian-group / zero-sum / cross-number formulation.
9. The attempted one-line cross-number closure failed; the failure must be preserved.

2. Global notation

Let:

$$R(X) := \sum_{x \in X} 1/x$$
$$\theta := 1 - 1/e$$

A signed reciprocal circuit is written as two sides:

$$P, Q \subseteq \{1, \dots, N\}, \quad R(P) = R(Q).$$

The circuit is primitive if there is no non-empty proper pair $P' \subseteq P$ and $Q' \subseteq Q$ such that $R(P') = R(Q')$. For fixed Q, write $S_Q := R(Q)$.

The fixed-Q programme aims to prove:

$$|P| \geq (\theta + \epsilon) N \text{ and } R(P) = S_Q$$

implies the circuit is not primitive.

This would be a serious fixed-negative-side obstruction. It would not, by itself, solve full Erdos #319.

3. Updated status table

Layer	Current status	New in this addendum	Distance to full #319
Concrete example $Q=(3,5,7,9,11,13,15,17)$	Strong numerical sanity check	Keep as example-level evidence only	Very far
Fixed-Q singleton gate	Strong architecture, not fully verified	Needs exact Liu-Sawhney parameter audit	Meaningful partial result if completed
Minimal-over-one fixed Q	Promising but technical	May feed into singleton-gate theorem	Partial
Arbitrary fixed Q	Closer but still open	Reduced to target-selection / zero-sum / cross-number	Medium-far
Prime-only fixed Q	Likely close conditional on reachability	σ_Q joint gate gives a cleaner route	Partial, plausible
Composite/mixed fixed Q	Main finite obstruction	Need prove/disprove target-selection lemma	Open
Growing Q_N	Long-term	Constants may decay; uniform estimates needed	Far
Full Erdos #319	Open	Needs two-sided spectrum/intersection theorem	Far

4. The Φ_q gate

Define $\Phi_q(u)$ as the minimum reciprocal mass of uN integers in $[1,N]$ avoiding multiples of q . The working formula is:

$$\Phi_q(u) = (1 - 1/q) * \log((1 - 1/q) / (1 - 1/q - u)).$$

Verification route: sort the nonmultiples of q in $[1,N]$; choose the largest uN such nonmultiples to minimize reciprocal mass; approximate by an integral; recover the displayed formula.

Status: essentially elementary/checkable, subject to finite- N error terms.

5. Fixed-Q singleton-gate theorem candidate

Let D in Q and suppose:

$$S_Q < \min_{p^a \mid\mid D} \Phi_{p^a}(\theta).$$

The intended theorem is: for every $\epsilon > 0$, for all sufficiently large N , there is no primitive circuit with fixed Q , arbitrary P , $|P| \geq (\theta + \epsilon)N$, and $R(P)=S_Q$.

Intended proof: Φ -gate forces positive local mass in P at every $p^a \mid\mid D$; Liu-Sawhney reachability produces $B \subseteq P$ with $R(B)=1/D$; then $B \cup \{D\}$ is a proper signed zero-subcircuit.

Status: strong proof architecture, not yet a fully verified theorem. The remaining verification is the Liu-Sawhney interface, pruning, exact target, and Archimedean window.

6. Dynamic single-prime gate

For a prime p , define $P_p = \{n \in P : p \text{ divides } n\}$ and $\alpha = p R(P_p)$. Since $|P_p| \leq N R(P_p) = \alpha N/p$, $P_{\text{not } p}$ has density at least $\theta + \epsilon - \alpha/p$.

$$S_Q - \alpha/p \geq \Phi_p(\theta + \epsilon - \alpha/p).$$

Thus if $S_Q < \Phi_p(\theta + \epsilon)$, one expects $\alpha = p R(P_p) \geq c_{\{p,Q,\epsilon\}} > 0$.

Status: promising lemma candidate. Needs rigorous monotonicity, error control, and finite-N handling.

7. Joint-avoidance / σ_Q gate

Define $\pi(Q) = \{p : p \text{ divides } q \text{ for some } q \in Q\}$ and $\sigma_Q = \prod_{p \in \pi(Q)} (1 - 1/p)$. Let P^* be the elements of P coprime to the product of primes in $\pi(Q)$. Then $|P^*| \leq \sigma_Q N + o(N)$.

$$|P| \leq \sigma_Q N + N \sum_{p \in \pi(Q)} R(P_p) + o(N).$$

Therefore, if $|P| \geq (\theta + \epsilon)N$, then $\sum_{p \in \pi(Q)} R(P_p) \geq \theta + \epsilon - \sigma_Q - o(1)$.

Status: looks like a genuinely provable fixed-Q lemma and is one of the strongest additions from the final messages.

8. Prime-only fixed Q

If $Q = \{p_1, \dots, p_k\}$ is prime-only and $S_Q > 1$, then $\pi(Q) = Q$. By the joint gate, some p_0 in Q should have $R(P_{\{p_0\}}) \geq c_{\{Q,\epsilon\}} > 0$.

Taking $Q' = \{p_0\}$ gives $R(Q') = 1/p_0$. Liu-Sawhney reachability, if applicable, should allow P to hit $1/p_0$, producing a proper subcircuit.

Status: likely close, conditional on reachability. Still not a proof until the Liu-Sawhney interface is verified.

9. Composite/mixed fixed Q: target-selection problem

For composite-only or mixed Q , the joint gate may force mass at primes in $\pi(Q)$ even when those primes are not elements of Q . Hence target $1/p$ may not be available.

$$T_Q := \{R(Q') : \emptyset \neq Q' \subseteq \pi(Q) \text{ and } Q' \not\subseteq Q\}.$$

The arbitrary fixed-Q target-selection problem is to find $Q' \subseteq \pi(Q)$ such that Liu-Sawhney-type reachability can force $B \subseteq P$ with $R(B) = R(Q')$. Then $B \cup Q'$ is a proper subcircuit.

Status: correct formulation for arbitrary fixed Q . Target selection is real, not cosmetic.

10. Obstruction group / zero-sum formulation

For primes outside the forced/bad-prime set, define a finite p -adic obstruction group G_Q and map each q in Q to a vector v_q in G_Q .

The condition that $R(Q')$ has denominator supported only on forced primes should correspond to $\sum_{q \in Q'} v_q = 0$ in G_Q .

If target selection fails, $(v_q)_{q \in Q}$ should be a minimal zero-sum sequence, or atom. Define $k((v_q)) = \sum 1/\text{ord}(v_q)$. Since $\text{ord}(v_q) \leq q$, $S_Q \leq k((v_q))$.

Status: correct framework, not yet a proof. The next algebraic task is a special bound for reciprocal-realizable atoms.

11. Failed one-line cross-number closure

The proposed closure G_Q cyclic, $K(G_Q)=1$, hence $S_Q \leq 1$ fails because $K(C_n)=1$ is false for composite cyclic groups in the needed maximal-cross-number sense.

Counterexample in C_6 : (2,2,3,5) is minimal zero-sum because $2+2+3+5 = 12 \equiv 0 \pmod 6$ and no proper subsequence sums to zero. The orders are 3,3,2,6, giving cross number $1/3+1/3+1/2+1/6 = 4/3 > 1$.

The valid conclusion is only $S_Q \leq k((v_q)) \leq K(G_Q)$. Thus target selection is forced if $S_Q > K(G_Q)$, not merely if $S_Q > 1$.

Status: one-line closure rejected. The cross-number route remains valuable, but the needed theorem is special to reciprocal-realizable atoms.

12. Current problem list

Problem A: Liu-Sawhney interface. Verify that the reachability theorem applies to target $R(Q')$ under our local-mass hypotheses.

Problem B: Composite target selection. Prove or disprove: if Q is fixed, $1 \notin Q$, and $S_Q > 1$, then there exists proper Q' whose target denominator uses only forced primes.

Problem C: Reciprocal atom cross-number bound. If target selection fails, the obstruction sequence is a minimal zero-sum atom. We need either no reciprocal-realizable atoms with $S_Q > 1$, or a weaker sufficient condition such as $S_Q > K(G_Q)$.

Problem D: Growing Q_N . Even if fixed Q closes, growing Q_N requires uniform bounds. Constants from joint gate and local mass may decay.

13. Updated honest conclusion

Arbitrary fixed Q is closer, but still open. Growing Q_N and full #319 remain long-term.

Fixed- Q gate theorem architecture is serious; prime-only fixed Q is likely close through the σ_Q joint gate modulo Liu-Sawhney reachability; composite/mixed fixed Q has been reduced to a target-selection / zero-sum / cross-number problem.

The one-line cross-number proof failed and should not be reused. The remaining algebraic task is to prove a special cross-number bound for reciprocal-realizable atoms, or classify counterexamples.

14. Wording for subsidy / public progress report

Recommended wording: We have not solved Erdos #319. The strongest new progress is a fixed-negative-side programme: a Φ -based gate theorem candidate, a joint-avoidance σ_Q local-mass lemma, and a reduction of arbitrary fixed Q to a finite target-selection problem. The latter now has a precise

zero-sum/cross-number formulation. A proposed one-line closure was rejected after finding a cyclic-group counterexample, so the remaining task is sharply localized: verify the Liu-Sawhney reachability interface and prove or classify the reciprocal-realizable atom obstruction. This is genuine progress, but not a completed theorem or a full solution.

Do not write: Solved Erdos #319; arbitrary fixed Q solved; dynamic gate theorem proved; or the cross-number closure works because $K(C_n)=1$.

15. Immediate next actions

Create a Liu-Sawhney parameter dictionary matching our notation to Proposition 5.2 and the pruning lemmas.

Prove the σ_Q gate cleanly. This is the lowest-risk new lemma and should be written first.

Run the prime-only fixed- Q test using the joint gate plus target $1/p_0$; isolate exactly what reachability statement is needed.

Implement a target-selection enumerator: for fixed Q , enumerate all non-empty proper Q' , reduce $R(Q')$, and classify denominator supports.

Build a cross-number obstruction notebook for target-selection failures.

Do not attack growing Q_N first; fixed Q has stable constants, while growing Q_N needs uniform estimates.